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What FiO2 range would expect a nasal cannula to deliver?	The FiO2 delivered by a nasal cannula can vary significantly, but the ranges you can predict that would be delivered by varying flow rates are: 1 L/min: 0.21-0.24, 2L/min: 0.23-0.28, 3L/min: 0.27-0.34, 4L/min: 0.31-0.38, 5 or 6 L/min: 0.32-0.44. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1282.*
What are the two main factors that determine the resistance to gas flow in an anesthesia circuit?	Resistance to gas flow is a function of the length of the tube and its diameter. The shorter the length and the larger the diameter, the less resistance it offers. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 111.*
What is meant by zero, low, medium, and high-capacity oxygen delivery devices?	The capacity of an oxygen delivery device refers to the size of the built-in oxygen reservoir. A nasal cannula is a zero-capacity device as it has no oxygen reservoir. A tracheostomy mask or pediatric face mask is considered a low capacity device. A simple face mask and an aerosol face tent are both considered medium capacity devices. A nonrebreather mask is a high capacity device. *Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 73-74.*
Why should cuff pressure be monitored in patients undergoing long-term ventilation via an endotracheal tube?	Cuff pressure that is too high can cause ischemia of the tracheal wall. Low cuff pressure in the endotracheal tube increases the risk that material can pass beside the cuff and into the lungs and is associated with an increased risk of pneumonia in patients on long-term ventilation. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 352-353.*
What are the concerns one may have when using an LMA?	An LMA doesn't protect against gastric secretions. Ventilation requiring pressures in excess of 20 cm H2O may result in inflation of the stomach. It can become malpositioned, resulting in an inability to ventilate. It is contraindicated in pharyngeal pathology such as tumor or abscess. Pathology at or below the level of the LMA may make it an ineffective means of ventilation. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 451.*
How do you choose the appropriate size LMA?	The LMA comes in several sizes: 1 for infants less than 5 kg, 1.5 for 5-10 kg, 2 for children 10-20 kg, 2.5 for children 20-30 kg, 3 for children and small adults between 30-50 kg, 4 for 50-70 kg, 5 for 70-100 kg, and 6 for patients greater than 100 kg. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 238.*

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Does an LMA protect the airway from pharyngeal secretions?	Yes. It does not, however, protect the airway from gastric secretions. Aspiration is a possibility in patients who experience reflux during the anesthetic. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 317.*
Will an LMA protect against laryngospasm?	No. The LMA is a supraglottic airway. Because it does not stent the cords open as an endotracheal tube would, it is unable to prevent laryngospasm. Additionally, a laryngeal mask airway normally produces a gas leak when air pressures reach 20 cm H2O and would be ineffective in producing the amount of positive pressure required to break a laryngospasm. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 556.*
What are the advantages and disadvantages of a low-volume, high-pressure cuff on an endotracheal tube?	A low-volume, high pressure cuff offers increased visibility during intubation because the cuff lies flat against the tube when not inflated. It also offers greater protection against aspiration. Because it requires a high intracuff pressure to overcome the decreased compliance of the cuff wall, it is difficult to assess how much pressure is being applied to the tracheal wall. Because this results in an increased risk for tracheal ischemia it is not suitable for long-term intubations. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 351.*
What is the indication for using an anode (armored) endotracheal tube?	The anode (or armored) tube contains an embedded wire that is designed to prevent kinking when the tube is bent. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 560.*
What are the differences between silicone, PVC, polyurethane, and red rubber endotracheal tubes?	Red rubber endotracheal tubes are opaque endotracheal tubes that can be cleaned and sterilized for reuse. They are more prone to kinking, occlusion by secretions, have high-pressure cuffs, and can be a trigger for patients with latex allergy. Polyvinyl chloride endotracheal tubes are transparent, soften when warmed, and are less prone to kinking than red rubber tubes. Silicone endotracheal tubes are more expensive than PVC, soft, and are reusable. Polyurethane tubes are a more recent adaptation. Studies show that endotracheal tubes with a polyurethane cuff have a decreased association with early postoperative pneumonia. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 336-337.*
What is the indication for using an oral or nasal RAE endotracheal tube?	Nasal and oral RAE tubes possess pre-formed bends that direct the tube away from the surgical field and are especially useful in oral and facial surgeries. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 560.*

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What is the indication for using a double-lumen endotracheal tube?	Double-lumen tubes and bronchial blocker tubes possess the ability to direct gas flow to one or another lung during surgery. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 560.*
What internal diameter endotracheal tube would be appropriate for a full-term infant?	For a full-term infant, a 3.5 mm internal diameter tube is recommended. For older children, the formula: 4 + (Age/4) is a useful predictor for the appropriate endotracheal tube size. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 321.*
What is the most reliable method of confirming placement of the endotracheal tube in the trachea?	Although several factors such as chest rise, oxygen saturation, direct visualization, and breath sounds should be assessed, persistent end-tidal carbon dioxide detection is the most reliable indicator that the endotracheal tube is positioned in the trachea. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 371.*
How should the endotracheal tube be positioned when performing a nasal intubation?	When performing a nasotracheal intubation, the tracheal tube should be inserted into the nares at an angle perpendicular to the face with the bevel directed away from the turbinates. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 330.*
What is the primary advantage of the video-assisted laryngoscope (ex. Glidescope)?	It can provide equal or better visualization of the airway without manipulation of the head into the sniffing position. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 457.*
What are the advantages and disadvantages of the Bullard laryngoscope?	The Bullard laryngoscope is quicker to use than a flexible, fiberoptic scope. It is useful in patients with limited neck or oropharyngeal mobility. It results in less dental trauma and has a lower risk of failed intubation than traditional laryngoscopy and is more resistant to problems with secretions. It does not accommodate double-lumen or metallic laser tubes well and using an andotracheal tube larger than 7.5 mm can result in posterior displacement of the introducing stylet. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 323.*

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What are some of the limitations of the traditional flexible fiberoptic laryngoscope? How can they be overcome?	The lens of the scope can become fogged, making visualization difficult. This is more common if the scope is cold, so soaking it in warm saline prior to the procedure helps prevent fogging. The fiberoptic strands in the scope are fragile and can become broken easily. As strands break, visualization can become limited. Care must be taken to protect the instrument from damage and it should be stored in a safe, secure location. Secretions and blood can obstruct the view of the scope. It contains a suction channel that runs the length of the scope that can be irrigated with normal saline to wash secretions out of the viewing area. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 454-455.*
What are the advantages of the use of a lighted stylet over traditional laryngoscopy?	Although the success rate in using a lighted stylet such as the Trachlite is similar to that of traditional laryngoscopy, it is less affected by an anterior airway, is less stimulating than laryngoscopy, and results in a lower incidence of sore throat. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 453.*
Is a manual resuscitator with a self-inflating bag a good choice for spontaneously breathing patients? Why or why not?	A manual resuscitator with a self-expanding bag is not the ideal choice for spontaneously breathing patients because if the patient cannot exert a sufficient negative pressure to open the inspiratory port they may inhale room air via the expiratory port. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 189.*
What are the components on all self-inflating manual resuscitators? What components are optional?	All self-inflating manual resuscitators have a self-expanding bag, a bag inlet valve, and a nonrebreathing valve. Optional components include CO2 detectors, airway pressure monitor, PEEP valve, pressure-limiting devices, and scavenging devices. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 183.*
What are the factors that determine the patient's minute volume when being ventilated by a self-inflating manual resuscitator?	The tidal volume, respiratory rate, and operator skill level are the determinants of the patient's minute volume when being ventilated by a self-inflating manual resuscitator. The operator must be able to maintain an open airway and adequate seal when using the device to deliver adequate minute ventilation. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 188.*
According to ASTM standards, what is the minimum FiO2 a self-inflating manual resuscitator should be able to deliver when connected to an oxygen source?	40% *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 188-189.*

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With what airway device is the risk of high airway pressure the greatest when ventilating with a self-inflating manual resuscitator?	High airway pressure is a hazard when using a self-inflating manual resuscitator if a pressure relief valve is not utilized. The risk is greater with endotracheal tubes than with mask or supraglottic devices such as the LMA or laryngeal tube which tend to leak at a lower pressure than an endotracheal tube. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 189.*
Which cuff should be inflated after insertion of a Combitube?	After inserting a Combitube, you should inflate both the proximal and distal cuffs. The Combitube will be placed within the esophagus in 95% of insertions in which case ventilation through the blue tube will force air through the side perforations and into the larynx. If the Combitube enters the trachea, ventilation can proceed by connecting the circuit to the shorter, clear tube. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 320.*
Which endotracheal tube stylet allows the stylet angle to be adjusted during laryngoscopy?	The Schroeder stylet is a plastic, disposable stylet that allows the stylet angle to be adjusted while performing the laryngoscopy. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia, PA: Elsevier Saunders; 2011: 231.*
What serves as confirmation of correct placement of an Eschmann stylet?	Confirmation of appropriate placement of an Eschmann stylet (gum elastic bougie) is made when you feel the stylet travel across the bumpy surface of the tracheal rings, but this sensation may not always be noticed. Although the stylet may strike a surface and stop, this can occur in both the esophagus and the trachea and is not confirmation. Striking a surface should be avoided if possible to prevent a bronchial or tracheal tear. When placed into the trachea, the stylet should be stopped at 25 cm at the lip and held in place. *Nagelhout JJ, Zaglaniczny KL. Nurse Anesthesia. 4th ed. Philadelphia, PA: WB Saunders Company; 2010: 454.*
What is the maximum recommended suction catheter sizes for endotracheal tubes sized 5.0 to 9.0?	The maximum suction catheter size for size 5.0-6.0 endotracheal tubes is 8 French. For a 7.0 ETT, it is 10 French. For an 8.0 ETT, it is 12 French. For a 9.0 ETT, the maximum size is 14 French. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 39.*
What are contraindications to the insertion of a nasopharyngeal airway?	Contraindications to the insertion of a nasopharyngeal airway include: anticoagulation, basilar skull fracture, history of nosebleed requiring medical treatment, and nasal or nasopharyngeal deformity. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 261.*

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